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Design and Application of Human resource management system Based on Data Mining Technology

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Abstract

In order to solve the problem of low efficiency of human resource management in the traditional mode, the author proposes a research on data mining technology. The author applies the technology of data mining to human resource management, and designs the employee performance appraisal algorithm based on the analytic hierarchy process, human resource flow algorithm based on Markov model, human resource management system based on ID3 decision tree employee satisfaction algorithm, and use actual data to test the system. The result shows, in the personnel composition prediction experiment, the system prediction error is less than 0.031, and the prediction accuracy rate is over 95%. The system can predict the future personnel flow of the enterprise; In the actual use experience, the satisfaction with the system is above 75%. Data mining methods are all based on examples, so each enterprise needs to use different methods to analyze the data, and then compare the analysis results, according to the analysis results, find out the method suitable for your enterprise in the follow-up human resource management work. The method adopted by the author integrates data mining into human resource management, it provides a reasonable reference for the changing trend of enterprise personnel structure, but there are systematic and complex problems in actual use, which need further improvement.

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Keywords: Human resource management; data mining; Markov; ID3 decision tree

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1. Introduction

In modern management, market capitalization is the capital required to operate a sector. Business capital is generally divided into two types: foreign capital and domestic capital [1]. A company's external capital can be divided into human capital, management capital, financial assets, information resources, technology capital, capital resources, internal environment, and business management.[2] A company's internal capital is divided into business assets, business assets, environmental capital, and business capital. Human capital is an important factor in business management. The sector's human resource management is generally divided into seven areas: recruitment, management training, organization, remuneration, job evaluation, and human resource planning and management [3].

In the era of globalization of information and information technology, human resource management can not adapt to the rapid development of the information industry. marketing, and the media have become an integral part of human governance [4]. Knowledge of human resource management is the use of computers and other information technology in conjunction with the management procedures of border human resources agencies to use information technology. for data collection and analysis in order to maximize the benefits of population distribution [5]. Human Resource Informatization is the use of computer-aided management tools, as well as the management of existing human resources, to address human resource management. This combination contributes to the success of the work manager. The process of introducing people management and the integration of management systems will help improve the performance of leaders, improve the efficiency of business management and reduce the cost of business management [6]. The English name for human resource management is Human Resource Management System (HRMS), which is a management system for communication technology and human management. The development of union and system began in 1969 and has undergone three generations of evolution from what it seems to be today. Early procedures used computers to calculate wages [7]. At the time, the system was restricted by counting the power of the computer and could not perform tasks such as collecting payment information, creating employee information, and analyzing payment information. As computer technology was rapidly developed in the 1980s. At that time, people were told about human resources to collect, use, and analyze payment information from data, which during the so-called second administration, during both contain historical inquiries. , human report and data analysis function. Designing these systems will eliminate the uncertainties of the previous process. In the 1990s, with the rapid development of computer technology, many improvements have been made to human administration. The second generation systems focus on data concentration, control centralization, and control centralization. The current process has improved the functionality of integrated data management, collaborative services, data analysis, and broadcast analysis [8-9].

Traditional human resource management generally adopts four management methods: Statistical technology, econometric model, survey technology and information technology [10].

(1) Statistical techniques are usually used for the comparison of quantitative data, this method is generally used to analyze and compare the data by calculating the probability of the data, the variance of the data, the mean of the data, and the total value of the data. For example: This method is generally used in the evaluation of employee salary system and employee attendance evaluation.

(2) Econometrics technology is based on the premise of economic theory, only through mathematical modeling technology can the relationship between variables be discovered. The regression algorithm is usually used for analysis and prediction. Survey technology is a systematic and scientific investigation of human resource management activities.

(3) Questionnaire survey is one of the most commonly used survey and analysis methods for human resource managers.

(4) Information system technology is a process in which human resource managers store and process data through computers. This method can improve the work efficiency of managers, and can also ensure that the data can be well stored electronically [11-12].

These four traditional human resource management techniques are currently the most commonly used techniques for human resource management. Although it has been widely used and achieved good practical results, there are still some defects. Due to the characteristics of human beings, and qualitative analysis is usually used in human resource management, there is a lack of quantitative analysis, so it is easy to introduce human subjective thinking, resulting in inaccurate human resource management analysis[13].

2. Literature review

Data mining was planned in the 1980s and 1990s and developed rapidly, making it a new kind of scientific research focused on the industry. As a new industry, data mining includes statistics, machine learning, data retention, mathematical statistics, database technology, image, data recovery, and more. Areas of concern include model recognition, spatial data analysis, intelligence (AI), graph theory, probability theory, image processing, and other fields [14].

From a visual point of view, data is simply the process of deleting an incomplete, noisy, inaccurate and random amount of data that is hidden inside it, which people do not know in advance, but will need. in the kitchen. These terms have multiple meanings: The information must be realistic, large, and loud; findings are the experience that a user needs; discovers knowledge to be acquired, understood and applied; Without delaying the findings, Global Knowledge only supports some of the publications.

From an industry perspective, data mining (DM) is the following process: Regarding the new way of data technology industry, the process of processing DM is create, delete, exchange, analyze, and model large business data files. . Important information on business decisions can be obtained from the results of market analysis. Data mining is the process of digging and examining data in depth. Now that industrial automation has been introduced in all areas of life, the pure industry has developed a lot of industry information in the industry, and after in-depth examination, contains important information and competition for the market. tau [15].

In short, the data simply means: the predetermined correlation goals of a business, the analysis and processing of large data on the market, the analysis of a variety of data hidden, unknown, ambiguous, random, or existing rules. an expert and effective tool for further modeling of accomplishments.

Human researchers are only able to identify data on the surface of data, and it is not possible to obtain internal and implicit relationships of data. and information. With the development of big data and cloud computing, data mining technology has been widely used in all stages of life, because the data industry has increased many years after the integration. written at all stages of life. With the help of data mining, we can transform this data into the knowledge and data we need. Data mining is the extraction of valuable data into data by some method of research including data, and such a process can be called data mining. More and more people are now studying data mining, and data from data mining has been successfully used in industry data analysis, industry analysis, industry data mining, industrial engineering, care engineering, and other industries. The purpose of human control is human, and these objects are complex and unstable. It is true that the process of always following a good assessment is not able to meet the needs of the leaders. With the advent of cloud technology in recent years, data mining has become less understood and used by humans, and its role is to find the secret code and compatibility in existing data and solutions, which is another technology. cannot be compared. Data mining technology is based on the concept of traditional arithmetic and expands and expands statistics naturally. Based on these advantages, data mining technology has been widely used in many industries, and significant benefits have been achieved by this technology. The advantage of integrating data mining into human resources is that high and complex mathematical analysis can be used by algorithms and then integrated into the system. Therefore, management does not require a deep understanding of this screening process, and is able to extract, analyze, and analyze the information they need from the system, and allows users to focus on the various issues they want. address.

3. Research methods

3.1. Data Mining

The main tasks of data mining are Clustering, Association analysis, Classification, Prediction model, Time-series pattern and Variance analysis.

Cluster analysis: Cluster analysis is to study the similarity between data, and divide the data into N categories through this similarity. The data in each category are similar data, and similarly, the data in different categories are dissimilar data. Cluster analysis can discover the distribution pattern between data and data, which is to establish the analysis of data on a macro level. Of course, through cluster analysis algorithm, it can also find out the hidden relationship between data and data.

Correlation analysis: It is information that is relevant. Organizational analysis is usually used to identify the relationship between two or more differences, and is called cohesive differences. Its purpose is to identify

confidential and non-confidential information. Relationships can be divided into three types: easy connection, temporary connection, and interconnected connection. Relationship analysis can use both the values of support and trust to measure the relationship.

Classification analysis: Classification models are generally represented by decision tree models and rules. He is mainly used to find out from a certain concept to describe a certain category. Prediction model: Through historical data, find out the change rule or change rule of these data, and build a model for the data through this rule. Use this model to predict future trends and characteristics.

Timing pattern: Through repeated practice, and then finding out the pattern with a higher frequency of occurrence and a higher probability of occurrence from these practices, such a process is called a sequential pattern. Time series models are similar to forecast models, both predicting future values based on existing data. The evolution process of data mining is shown in Table 1.

Table 1 Evolution process of data mining

Evolutionary stage	Business problem	Support technology	Features
Data Search (1960s)	"What was my total income in the first 10 years?"	Computers, cassettes and disks	Provide historical, static data
Data Access (1980s)	"What was the profit of the division in New York last May?"	Relational Database (RDBMS), Structured Query Language (SQL)	Provides record-level historical, non-static data
Data Warehousing, Decision Support (1990s)	"What was the profit of the division in New York in March last year? What conclusions will other division cities draw from this?"	Online Analytical Processing (OLAP), Multidimensional Databases, Data Warehousing	Provides multiple levels of backtracking, non-static data
data mining	"What are the sales trends in New York next year? Why?"	Data mining algorithms, multi-core computers, massive data warehouses	Provide predictive information

3.2. Performance Appraisal Model

One of the important tasks in human resource management is performance appraisal, which is a comprehensive evaluation of employees' work performance, character, attitude, and ability based on specific standards and scientific methods. The status of performance management in human resource management cannot be ignored, and its importance has been recognized by the vast majority of enterprise managers. In order to achieve the purpose of performance management, enterprises must pass performance appraisal, an important management tool and management method, so as to achieve the effect of improving and enhancing work performance. Although the importance of performance appraisal has been recognized by most enterprises, few enterprises have established effective, scientific and reasonable performance appraisal. The main reason for this is that there are many difficulties in performance appraisal in practice. The main difficulty is how to make a comprehensive quantitative evaluation of employees' work goals, performance goals, and management priorities, so that the company's evaluation system can be fair, comprehensive and complete.

Enterprises usually conduct performance appraisal of employees through the AHP model, which can make the appraisal and evaluation more observable and scientific. In actual work, it is possible to evaluate employees who have performed poorly for several consecutive years, or have a downward trend in assessments for several consecutive years, or stagnant employees, take corresponding organizational measures to promote the enthusiasm of employees, promote the improvement of their comprehensive assessment scores, so as to achieve scientific and reasonable assessment of employees.

3.3. Analysis of Human Resource Flow

The flow of human resources in enterprises is mainly divided into two categories: Normal flow and abnormal flow. Normal flow refers to the natural flow of employees due to birth, old age, sickness and death, or the flow of job changes due to employee promotion or job demotion. The abnormal flow of human resources can be dismissed

by the company due to limited personal ability, limited technical ability, lack of knowledge, and poor teamwork ability. This kind of loss also becomes active loss. According to the flow of human resources and the actual needs of the enterprise, accurately forecast the needs of various talents, reasonable introduction of talents has become an important factor restricting enterprises to maintain their advantages in the fierce market. Traditional human resource management methods are difficult to meet the needs of enterprises to predict the flow of human resources.

A Markov Model is a predictive model. In the model, the sequence of time is regarded as a random process, in this process, research the probability of the initial state of things in different states at various time points and the transition probability between states, and use this model to predict the state at a certain moment in the future. Markov processes can predict the future state of things based on the current state of events. After introducing the Markov model, you first need to understand the three conceptual words "state", "state transition" and "Markov process".

"State": Refers to a certain outcome of an event at a certain time or period. Generally speaking, the events that have been studied and the change of the predicted target have different divisions of the state.

◇ "State Transition": In the process of event occurrence, the transition from one state to another is called state transition. The state transition in which the event changes with the change of time becomes the state transition process.

◇ "Markov process": The Markov process is based on "no aftereffect". "No aftereffect" means that every time a thing happens, its state is only related to the state of the previous moment, and has nothing to do with the state of other moments in its historical process, or the process of state transition is independent, which This transition is called a Markov process.

Let X be the state of things at the moment, in order to facilitate the study, we discretize the continuously changing time; At the same time, it is necessary to divide the state of things into several levels, which correspond to the state space respectively. Mathematically expressed as: For any integer n , as well as $i, j, ik (k = 0, 1, 2, \dots, n - 1)$, there is formula (1):

$$P\{X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0\} = P\{X_{n+1} = j \mid X_n = i\} \quad (1)$$

If the conditional probability formula 1 is independent of n , the Markov chain has "stationarity" and can be expressed as formula 2. "Stationarity" means that the characteristics of the transition matrix P do not change over time. P_{ij}^1 can represent the probability that state i is transferred to state j after one step, or denoted as P_{ij} , then P_{ij} satisfies the condition in Equation 3.

$$P\{X_{n+1} = j \mid X_n = i\} = \dots = P\{X = j \mid X_0 = i_0\} = P_{ij}^1, (n = 1, 2, \dots) \quad (2)$$

$$\begin{cases} P_{ij} \geq 0 (i, j \in E) \\ \sum_{j \in E} P_{ij} = 1 (i \in E) \end{cases} \quad (3)$$

In the process of an event, the probability of starting from one state and transitioning to another state at the next moment is called the state transition probability, here, the matrix $P = (P_{ij})$ with P_{ij} as the element is called the state transition probability matrix, such as formula 4, in the formula, P_{ij} still conforms to the properties of formula 3.

$$P = \begin{bmatrix} P_{11} & P_{12} & \dots & P_{1n} \\ P_{12} & P_{22} & \dots & P_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ P_{n1} & P_{n2} & \dots & P_{nn} \end{bmatrix} \quad (4)$$

Since the Markov process needs to be based on "stationarity" and "no aftereffect", equation (5) can be obtained

$$p^k = p^{k-1} \cdot p = p^{k-2} \cdot p^2 = \dots = p^k \quad (5)$$

Here P^k is the probability matrix of the state of things going through the steps. After introducing the basic concepts, the next step is to start building a Markov prediction model. Markov's model is based on "no aftereffect" and "stationarity". Assuming that a thing has n mutually incompatible states, its initial state is $S^0P = [S_1^0, S_2^0, \dots, S_n^0]$, and $S_t^t (t = 1, 2, \dots, n)$ in the formula is the initial probability of being in the state, assuming that the probability of being in stage i after k -step transition is S_i^k , the equation (6) can be obtained from the Chepman-Kolmogorov (C-K) equation

$$S_j^{k+1} = \sum S_i^k \cdot P_{ij}, (j = 1, 2, \dots, n) \quad (6)$$

Equation 6 can be rewritten in matrix form to obtain Equation 7:

$$[S_1^{k+1} S_2^{k+1} \dots S_n^{k+1}] = [S_1^k S_2^k \dots S_n^k] \cdot \begin{bmatrix} P_{11} & P_{12} & \dots & P_{1n} \\ P_{21} & P_{22} & \dots & P_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ P_{n1} & P_{n2} & \dots & P_{nn} \end{bmatrix} \quad (7)$$

Using the recurrence relation, Equation (8) can be obtained:

$$\begin{aligned} S^1 &= S^0 \cdot P \\ S^2 &= S^1 \cdot P = S^0 \cdot P^2 \\ &\dots \\ S^{k+1} &= S^k \cdot P = \dots S^0 \cdot P^{k+1} \end{aligned} \quad (8)$$

Equation 8 is the Markov prediction model of the $(k + 1)$ period. From this formula, we can see that for the Markov chain, its state at any state moment is determined by the initial state and the probability of going through a state transition.

3.4. Analysis of employee satisfaction

The most intuitive response to employees' evaluation of an enterprise is through satisfaction, and the degree of satisfaction directly affects employees' loyalty and sense of belonging to the enterprise. In an enterprise, employees have expectations for the working environment provided by the enterprise, the management status of the enterprise, and the remuneration package. If these expectations are met, it affects the performance of employees. From different perspectives, to study employee satisfaction, we can get different indicators that affect employee satisfaction. Research on employee satisfaction indicators can start from the following directions: From the perspective of organizational behavior, from the perspective of employee personality, research on knowledge workers. From the perspective of organizational behavior, the indicators of employee satisfaction are as follows: Working hours, working environment, sense of achievement, salary status, corporate image, promotion opportunities, personal interests, and the relationship between leaders and colleagues.

Decision Tree is a simple and effective fast classification algorithm commonly used in data mining, this algorithm is an inductive learning algorithm based on data, which can be applied to data classification and data prediction. The algorithm starts from a set of unordered and irregular data, and infers the classification rules of the decision tree based on these data. The algorithm of the decision tree starts from the root node, and then descends to the child nodes in sequence, and performs attribute judgment and comparison at each internal node, selects the next branch according to the result of the judgment and comparison, and then prunes the decision tree after selecting the branch, the decision conclusion of the whole system will be obtained on the last child node of the decision tree. Therefore, from the root node to the child node corresponding to a conjunction rule, the entire tree corresponds to a set of disjunctive expression rules. There are many algorithms based on decision trees, and the ID3 decision tree algorithm is one of the most effective, fastest and simplest. The ID3 decision tree algorithm is a decision tree algorithm based on information entropy (Entropy), and it is also a decision tree algorithm proposed earlier and commonly used.

From a mathematical point of view, the algorithm model is as follows: Suppose the training data set is D_k , its attribute set $X = \{x_1, x_2, \dots, x_i, \dots, x_n\}$, randomly select an attribute x_i from the attribute set, through the S values of the value range $V(x_i) = \{v_1, v_2, \dots, v_{il}, \dots, v_n\}$ of the attribute of the x_i , divide D_k into S subsets $\{D_{k1}^i, D_{k2}^i, \dots, D_{kl}^i, \dots, D_{ks}^i\}$. The value of the attribute x_i of all instances in the sub-set D_{ks}^i is v_j . Here, the result table of D_k instance classification is $C = \{c_1, c_2, \dots, c_i, \dots, c_m\}$, if the actual number of subset D_{ks}^i is e_{ky} classification result c_j , the instance tree is $e_{ky}(c_j)$, $1 \leq j \leq m$, then the actual classification result of instance set D_{ks}^i is c_j and the probability is $P_j = \frac{e_{ky}(c_j)}{e_{ky}}$. The information entropy of the training set D_{kx} is shown in formula (9).

$$I(D_k^i) = \sum_{j=1}^m P_j \log_2 P_j = - \sum_{j=1}^m \frac{e_k(c_j)}{c_j} \log_2 \frac{e_k(c_j)}{c_j} \quad (9)$$

The heuristic function Entropy (D_k, x_i) is introduced to generate the ID3 decision tree, if we choose x_i to divide the training instance set D_k into S subsets, the information entropy (D_k, x_i) of each subset is expressed as the average information content of the training instance as in Eq. (10). We use heuristic rules to divide the subsets: after the training instance set D_k is divided into several subsets by any attribute x_i , we need to find the minimum entropy

of an instance, which is $\min\{I(D_k, x_i), x_i \in X\}$. Use this attribute-partitioned instance set to get the nodes of the decision tree.

$$I(D_k, x_i) = \sum_{y=1}^s \frac{e_k}{e_{ky}} I(D_k^i) \quad (10)$$

3.5. System Architecture Design

The core software architecture of the system is the C / S architecture, which consists of two parts software, Client and server. The use of C / S is based on high data rate, high security, and ease of use of the equipment. These measures can be tailored to the needs of the human resource management organization. The design of a comprehensive system for managing human resources is shown in Figure 1. The main basis includes 7 functional areas: personnel management, process management, work, interest management, business management, supervision, attendance management, and publishing statistics. Each major module has many different features.

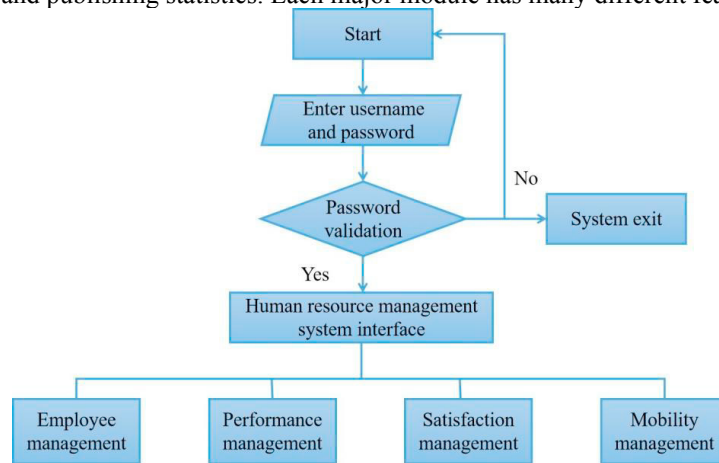


Figure 1 Functional modules of human resource management system

The login module of the human resource management system must be used by system administrators, staff file management staff, staff performance appraisal staff, staff satisfaction analysts, and staff flow forecast analysts. The specific system function design is shown in Figure 2.

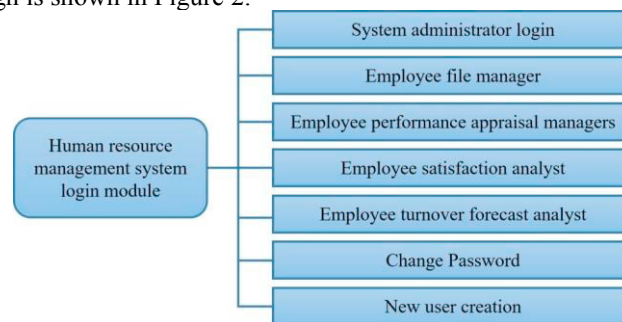


Figure 2 System login module design

The employee file management module mainly provides enterprises to manage the basic information of employees, the function of this module is shown in Figure 3. Through this module, users can enter basic employee information, modify existing employee information, and delete employee information. In this module, users can also operate employee onboarding, employee regularization, employee transfer, employee reward, employee punishment, employee reemployment, employee resignation, employee reinstatement, and employee promotion. At the same

time, the system can import and export the basic information of employees, and can produce employee resume forms.

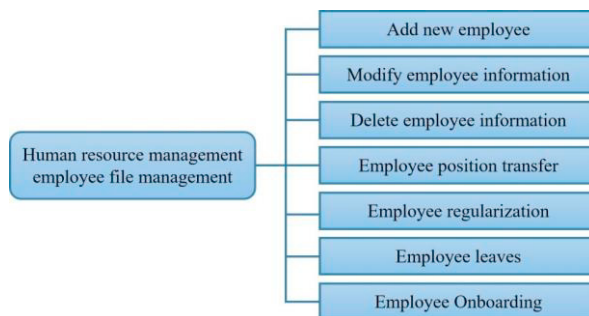


Figure 3 Employee file management module design

The employee performance appraisal module mainly provides enterprises to manage employee performance appraisal. The functions of this module are shown in Figure 4. Users can predict the performance appraisal of employees through the performance appraisal model based on analytic hierarchy process.

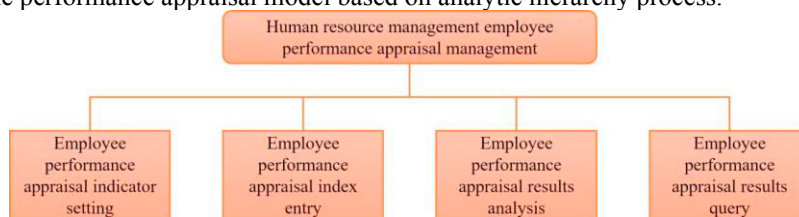


Figure 4 Employee performance appraisal module design

The employee satisfaction management module is mainly used by enterprises to manage employee satisfaction, the functions of this module are shown in Figure 5. The user can set the employee satisfaction index through this module, and then enter the satisfaction index.

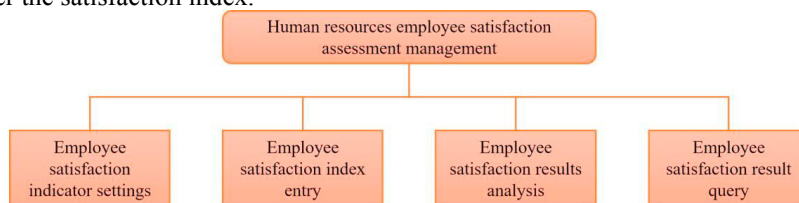


Figure 5 Employee satisfaction module design

Employee mobility analysis module mainly provides enterprises to analyze and manage employee mobility analysis, the function of this module is shown in Figure 6. The user can set the employee's flow analysis index through this module, and then enter the flow analysis index. After the entry is completed, the flow of employees is predicted and analyzed by the Markov algorithm.

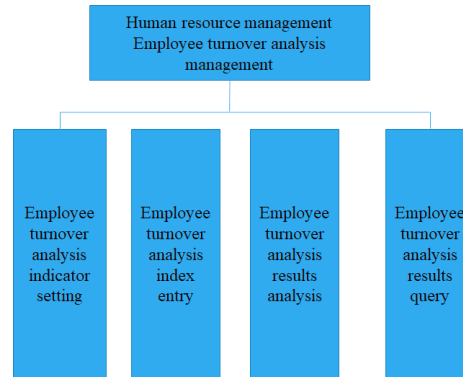


Figure 6 Employee Mobility Analysis Module

In this system, the operation of data is carried out through XPO. XPO builds a bridge between relational databases and real objects. It transforms operations on the database into operations on objects. This allows developers to pay more attention to the business logic of the entire system, rather than complex and cumbersome databases. The system is described by means of an E-R diagram as shown in Figure 7.



Figure 7 System E-R Diagram

4. Analysis of results

4.1. Prediction of personnel composition

The system prediction method is to train through multiple groups of historical data, and then predict the value at a certain time in the future, and then treat the predicted data as historical data to continue to predict, and cycle in turn to gradually predict the value of a future period. The data of personnel composition of multiple companies is used as the input of the network, and the obtained predicted value is used as the output of the network. And so on, so that you get the forecast values of other companies. The simulation results are shown in Figure 8 and Table 2:

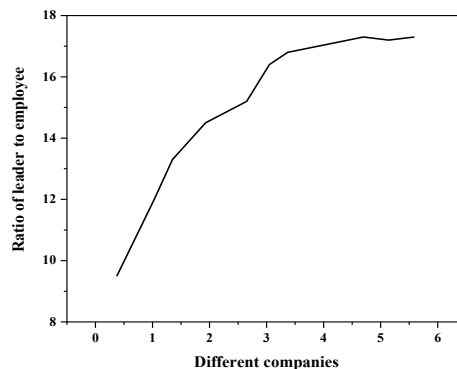


Figure 8 Comparison of actual and predicted values

Table 2 Comparison and error between actual and predicted values

Different companies												
Actual value	9.7	12	13.5	14.6	15.3	16.22	16.85	17.8	17.28	17.23	17.27	17.33
Predictive value	9.78	12.09	13.41	14.31	15.07	16.30	17.04	17.24	17.28	17.38	17.50	17.57
Error	0.008	0.007	0.006	0.019	0.015	0.005	0.011	0.031	0.042	0.013	0.013	0.014

As mentioned above, compared with the actual value, the error of the value predicted by the system is less than 0.031, and the prediction accuracy rate is more than 95%.

4.2. Human Resource Flow Forecast

The author analyzes the changes in the flow of human resources in a labor-intensive manufacturing enterprise, in order to make the enterprise develop better, it is necessary to predict the supply of human resources within the enterprise. The enterprise is divided into five categories of personnel: Staff, director, section chief, division chief, and manager. We classify the status of personnel, and they can be divided into the following six statuses: Staff, director, section chief, division chief, manager, and resignation. We selected the company's data, analyzed the existing data, and predicted the flow of human resources in the next year, as shown in Figure 9.

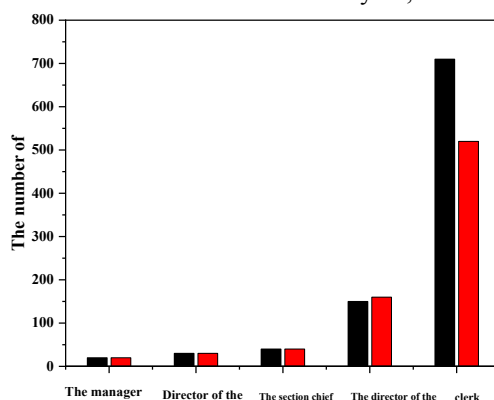


Figure 9 Prediction of personnel changes in an enterprise

As shown in the figure above, the personnel changes of the company in the next year are mainly due to the large changes in the number of employees, it is necessary to pay attention to recruiting such personnel in a timely manner.

4.3. System Satisfaction

Recommend the personnel management system designed by the author to multiple enterprises for practical use, and collect data and the experience of actual users, and obtain the system satisfaction as follows:

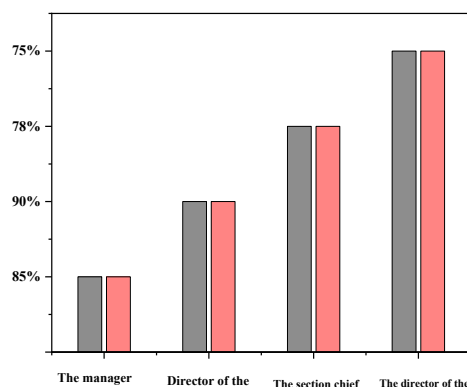


Figure 10 Satisfaction situation

As can be seen from Figure 10, in the actual use experience, the satisfaction with the system is above 70%. The specific collection of dissatisfied opinions is roughly as follows: The system is too complicated and it is cumbersome to use; among the users, the staff has a low level of education, and there are obstacles in using it, and there are many people who have not used it.

5. Conclusion

With the development of computer level, in many enterprises with a relatively deep degree of informatization, integrated business management system has become the main management idea to guide production and human resource management. Although many enterprises have adopted information-based human resource management, the existing human resource management system can only analyze the surface information of the data, and cannot obtain the internal and hidden relationship between the data and the data at a deep level. Influenced by the industry system, advanced human system management ideas cannot be popularized in enterprises. With the rise of big data and cloud computing, various industries have developed a wide range of applied data mining technologies. How to combine data mining technology and apply it to human resource management work, discover hidden laws in data, and effectively mine data in Valuable information.

The author, from the concept of human resources, starting from the insufficiency of traditional human resources, this paper discusses the development direction of human resource management. And based on this, the demand of human resource management system is analyzed. The author also applies data mining methods to human resource management, and integrates these analysis modules into human resource management system to form a set of intelligent human resource management system.

The research results mainly include:

- (1) Introduce the application of AHP in human resource management, its basic idea is to analyze, compare, and judge complex problems, arrange the weights of different elements, and finally make a final ranking by weighting, and use this method to solve the daily performance appraisal management of enterprise employees.
- (2) Introduces the application of Markov algorithm in human resource management, discretizes continuous time, the algorithm uses the current state to predict the future state given the current knowledge information.
- (3) The ID3 decision tree algorithm is used to analyze the satisfaction of employees, and by introducing the ID3 algorithm into the employee satisfaction survey, build a decision tree that affects employee satisfaction.
- (4) Establish a human resource management system based on data mining. The system includes AHP-based performance measurement, Markov-based human resource adequacy, and ID3 wood-based decision-making to customer satisfaction. Design is the design process, design of each function module, access to the system module, personnel data management, personnel performance measurement, people working staff control module, staff willing to analyze module and create database.

References

- [1] Boutmaghzoute, H. , & Moustaghfir, K. . (2021). Exploring the relationship between corporate social responsibility actions and employee retention: a human resource management perspective. *Human Systems Management*(12), 1-13.
- [2] Siddiqi, Z. , Mirani, M. A. , Nasim, S. , Shamshir, M. , & Nisar, S. . (2021). The relationship between human resource management and corporate social responsibility: a critical review. *Xinan Jiaotong Daxue Xuebao/Journal of Southwest Jiaotong University*, 56(2), 176-197.
- [3] Al-Swidi, A. K. , Gelaidan, H. , & Saleh, R. M. . (2021). The joint impact of green human resource management, leadership and organizational culture on employees' green behaviour and organisational environmental performance. *Journal of Cleaner Production*(6), 128112.
- [4] Yang, H. J. , Jeong, S. J. , & Yoon, S. W. . (2020). Enhancement for human resource management in the uld build-up process of air-cargo terminal: a strategic linkage approach. *Journal of Heuristics*, 26(2), 301-333.
- [5] Abuhasel, K. A. , & Khan, M. A. . (2020). A secure industrial internet of things (iiot) framework for resource management in smart manufacturing. *IEEE Access*, PP(99), 1-1.
- [6] Abdulraheem, A. S. , Zeebaree, S. , & Abdulazeez, A. M. . (2020). Design and implementation of electronic human resource management system for duhok polytechnic university. *Technology Reports of Kansai University*, 62(4), 1407-1420.
- [7] Xu, D. , & Xiao, X. . (2020). Influence of the development of vr technology on enterprise human resource management in the era of artificial intelligence. *IEEE Access*, PP(99), 1-1.
- [8] Oehlhorn, C. E. , Maier, C. , Laumer, S. , & Weitzel, T. . (2020). Human resource management and its impact on strategic business-it alignment: a literature review and avenues for future research. *The Journal of Strategic Information Systems*, 29(4), 101641.
- [9] Kim, T. H. , Kumar, G. , Saha, Rai, M. K. , & Alazab, M. . (2020). A privacy preserving distributed ledger framework for global human resource record management: the blockchain aspect. *IEEE Access*, PP(99), 1-1.
- [10] Numada, M. . (2021). Development of matching modeling for human resource allocation of shelter management by the set theory. *Journal of Disaster Research*, 16(4), 719-732.
- [11] Wang, W. , & Srivastava, G. . (2020). Enterprise human resource quality management model based on grey relational analysis. *International Journal of Performability Engineering*, 16(3), 419.
- [12] Wang, R. . (2021). Exploration of data mining algorithms of an online learning behaviour log based on cloud computing. *International Journal of Continuing Engineering Education and Life-Long Learning*, 31(3), 371.
- [13] Chowdhury, S., Dey, P., Joel-Edgar, S., Bhattacharya, S., Rodriguez-Espindola, O., Abadie, A., & Truong, L. (2023). Unlocking the value of artificial intelligence in human resource management through AI capability framework. *Human Resource Management Review*, 33(1), 100899.
- [14] Da Silva, L. B. P., Soltovski, R., Pontes, J., Treinta, F. T., Leitão, P., Mosconi, E., ... & Yoshino, R. T. (2022). Human resources management 4.0: Literature review and trends. *Computers & Industrial Engineering*, 168, 108111.
- [15] Alzoubi, H. M., Ghazal, T. M., Sahawneh, N., & Al-kassem, A. H. (2022). Fuzzy assisted human resource management for supply chain management issues. *Annals of Operations Research*.
- [16] Karatepe, O. M., Hsieh, H., & Aboramadan, M. (2022). The effects of green human resource management and perceived organizational support for the environment on green and non-green hotel employee outcomes. *International Journal of Hospitality Management*, 103, 103202.